Hyunbin Kim

Curriculum Vitae

Korea Research Institute of Bioscience and Biotechnology
(+82)10-9197-2561
khb7840@gmail.com
hyunbinkim.com
Github

Research Experience

2026.09– **Research Scientist**, *Korea Research Institute of Bioscience and Biotechnology (KRIBB)*, Daejeon, South Korea.

2025–2026.08 **Postdoctoral Fellow, Research Institute for Basic Sciences**, *Seoul National University*, Seoul, South Korea.

Principal Investigator **Dr. Martin Steinegger**, *Associate Professor, School of Biological Sciences*, Seoul National University, South Korea

2017 – 2020 **Researcher, Bioinformatics Analysis Team**, *National Cancer Center*, Goyang, South Korea, (Substitution for national service as a technical research personnel).

Principal Investigator **Dr. Dongwan Hong**, *Professor, College of Medicine*, Catholic University of Korea (*Lab web-page*)

Education

2021–2025 **PhD, Interdisciplinary Program in Bioinformatics**, *Seoul National University*, Seoul, South Korea.
Protein structure, Protein structure compression, Structural motif search, Protein structure prediction, *De novo* protein design, Development of scalable bioinformatics methods

Advisor **Dr. Martin Steinegger**, *Associate Professor, School of Biological Sciences*, Seoul National University, South Korea (*Lab web-page*)

2015–2017 **Master of Science in Agricultural Genomics, Interdisciplinary Program in Agricultural Genomics**, *Seoul National University*, Seoul, South Korea.
Protein-protein interaction network, Fungal plant pathology, Co-expression network

Advisor **Dr. Yong-Hwan Lee**, *Professor, Department of Agricultural Biotechnology*, Seoul National University, South Korea (*Lab web-page*)

2010–2015 **Bachelor of Science in Biological Sciences, School of Biological Sciences**, *Seoul National University*, Seoul, South Korea.

Software

2025 **Folddisco**, *Written in Rust*, A fast and scalable structural motif search tool, Repository.

2023 **Foldcomp**, *Written in C++*, Structure compression with torsion angles, Repository.

2019 **FIREVAT**, *Written in R*, A variant filtering tool using mutational signature profile, Repository.

Computer skills

Languages **C++, Rust, R, Python,** C, Go, Shell script

Web HTML 5, Javascript, Typescript

Design Illustrator, Figma, Affinity designer

Publications

First author

2026 **Hyunbin Kim**, Rachel Seongeun Kim, Milot Mirdita, Jaewon Yoon, and Martin Steinegger. Structural motif search across the protein universe with Folddisco. *Nature Biotechnology*, Jun 2026.

2026 **Hyunbin Kim**, Kyeongchae Cheong, Jongbum Jeon, Gobong Choi, Jaeho Koh, Hyeunjeong Song, Yoeguang Hue, Yebin Nam, Byungheon Choi, You-Jin Lim, Jaeyoung Choi, Ki-Tae Kim, and Yong-Hwan Lee. Magnet: Computational methods for constructing high-confidence protein-protein interaction networks in magnaporthe oryzae. *bioRxiv*, pages 2026–05. Cold Spring Harbor Laboratory, 2026.

2023 **Hyunbin Kim**, Milot Mirdita, and Martin Steinegger. Foldcomp: a library and format for compressing and indexing large protein structure sets. *Bioinformatics*, volume 39, page btad153. Oxford University Press, 2023.

2019 **Hyunbin Kim**, **Andy Jinseok Lee**, Jongkeun Lee, Hyonho Chun, Young Seok Ju, and Dongwan Hong. Firevat: finding reliable variants without artifacts in human cancer samples using etiologically relevant mutational signatures. *Genome Medicine*, volume 11, pages 1–17. Springer, 2019.

Co-author

2026 Yewon Han, Maxim I. Tsenkov, Niccolò A. E. Venanzi, Sooyoung Cha, Nilkanth Patel, Sreenath Nair, Razan Abbara, Damian Bertoni, Alejandro Chacón, Nick Dietrich, Boris Fomitchev, Yonathan Goldtzvik, Darren Hsu, Jeannie Austin, Joseph Ellaway, Kieran Didi, Gyuri Kim, **Hyunbin Kim**, Oleg Kovalevskiy, Dariusz Lasecki, Agata Laydon, Micha Livne, Paulyna Magaña, Maciej Majewski, Urmila Paramval, Risha Patel, Ivanna Pidruchna, Brianda Santini Lopez, Prashant Sohani, Ahsan Tanweer, Duc Tran, Kyle Tretina, Melanie Vollmar, Quan Vu, Augustin Žídek, Sameer Velankar, Martin Steinegger, Jennifer Fleming, Milot Mirdita, and Christian Dallago. Alphafold database expands to proteome-scale quaternary structures. *bioRxiv*, pages 2026–03. Cold Spring Harbor Laboratory, 2026.

2025 Jingi Yeo, Yewon Han, Nicola Bordin, Andy M Lau, Shaun Mathew Kandathil, **Hyunbin Kim**, Eli Levy Karin, Milot Mirdita, David Tudor Jones, Christine Orengo, and Martin Steinegger. Metagenomic-scale analysis of the predicted protein structure universe. *bioRxiv*. Cold Spring Harbor Laboratory, 2025.

2025 Hyunuk Eom, Sukhwan Park, Kye Soo Cho, Jihyeon Lee, **Hyunbin Kim**, Stephanie Kim, Jinsol Yang, Young-Hyun Han, Juyong Lee, Chaok Seok, Myeong Sup Lee, Woon Ju Song, and Martin Steinegger. Discovery of highly active kynureninases for cancer immunotherapy through protein language model. *Nucleic Acids Research*, volume 53, page gkae1245. Oxford University Press, 2025.

2024 Gyuri Kim, Sewon Lee, Eli Levy Karin, **Hyunbin Kim**, Yoshitaka Moriwaki, Sergey Ovchinnikov, Martin Steinegger, and Milot Mirdita. Easy and accurate protein structure prediction using colabfold. *Nature Protocols*, pages 1–23. Nature Publishing Group UK London, 2024.

2023 Caroline Puente-Lelievre, Ashar J Malik, Jordan Douglas, David Ascher, Matthew Baker, Jane Allison, Anthony Poole, Daniel Lundin, Matthew Fullmer, Remco Bouckert, **Hyunbin Kim**, Martin Steinegger, and Nicholas Matzke. Tertiary-interaction characters enable fast, model-based structural phylogenetics beyond the twilight zone. *bioRxiv*, pages 2023–12. Cold Spring Harbor Laboratory, 2023.

2020 Seungill Kim, Kyeongchae Cheong, Jieun Park, Myung-Shin Kim, Jihyun Kim, Min-Ki Seo, Geun Young Chae, Min Jeong Jang, Hyunggon Mang, Sun-Ho Kwon, Yong-Min Kim, Namjin Koo, Cheol Woo Min, Kwang-Soo Kim, Nuri Oh, Ki-Tae Kim, Jongbum Jeon, **Hyunbin Kim**, Yoon-Young Lee, Kee Hoon Sohn, Honour C. McCann, Sang-Kyu Ye, Sun Tae Kim, Kyung-Soon Park, Yong-Hwan Lee, and Doil Choi. Tgfam-finder: a novel solution for target-gene family annotation in plants. *New Phytologist*, volume 227, pages 1568–1581. Wiley Online Library, 2020.

2019 Ki-Tae Kim, Jaeho Ko, Hyeunjeong Song, Gobong Choi, **Hyunbin Kim**, Jongbum Jeon, Kyeongchae Cheong, Seogchan Kang, and Yong-Hwan Lee. Evolution of the genes encoding effector candidates within multiple pathotypes of magnaporthe oryzae. *Frontiers in Microbiology*, volume 10, page 2575. Frontiers Media SA, 2019.

2017 Seungill Kim, Jieun Park, Seon-In Yeom, Yong-Min Kim, Eunyoung Seo, Ki-Tae Kim, Myung-Shin Kim, Je Min Lee, Kyeongchae Cheong, Ho-Sub Shin, Saet-Byul Kim, Koeun Han, Jundae Lee, Minkyu Park, Hyun-Ah Lee, Hye-Young Lee, Youngsill Lee, Soohyun Oh, Joo Hyun Lee, Eunhye Choi, Eunbi Choi, So Eui Lee, Jongbum Jeon, **Hyunbin Kim**, Gobong Choi, Hyeunjeong Song, JunKi Lee, Sang-Choon Lee, Jin-Kyung Kwon, Hea-Young Lee, Namjin Koo, Yunji Hong, Ryan W Kim, Won-Hee Kang, Jin Hoe Huh, Byoung-Cheorl Kang, Tae-Jin Yang, Yong-Hwan Lee, Jeffrey L Bennetzen, and Doil Choi. New reference genome sequences of hot pepper reveal the massive evolution of plant disease-resistance genes by retroduplication. *Genome biology*, volume 18, pages 1–11. Springer, 2017.

2017 Jongbum Jeon, Ki-Tae Kim, Hyeunjeong Song, Gir-Won Lee, Kyeongchae Cheong, **Hyunbin Kim**, Gobong Choi, Yong-Hwan Lee, Jane E Stewart, Ned B Klopfenstein, et al. Draft genome sequence of the fungus associated with oak wilt mortality in south korea, raffaelea quercus-mongolicae kacc44405. *Genome announcements*, volume 5, pages 10–1128. American Society for Microbiology 1752 N St., NW, Washington, DC, 2017.

Presentations

Oral presentations

2026 **Hyunbin Kim**. Structural motif search across the protein universe with folddisco. In *LG AI Research Seminar, Seoul, South Korea*. LG AI Research, 2026.

2024 **Hyunbin Kim**, Rachel Seongeun Kim, Milot Mirdita, and Martin Steinegger. Structural motif detection on the scale of the protein universe with folddisco. In *APBJC2024, 1st Asia&Pacific Bioinformatics Joint Conference JSBi, GIW, InCoB, APBC, and ISCB-Asia*. ISCB, 2024.

2023 **Hyunbin Kim**, Milot Mirdita, and Martin Steinegger. Foldcomp: scalable solution for compressing huge protein structure database. In *ISMB/ECCB 2023, The 31st Annual Intelligent Systems For Molecular Biology and the 22nd Annual European Conference on Computational Biology, Lyon, France*. ISCB, 2023.

2017 **Hyunbin Kim**, K. Cheong, K.T. Kim, J. Jeon, G. Choi, and Y.H. Lee. Magnet: the integrated gene network of the rice blast fungus *Magnaporthe oryzae*. In *29th Fungal Genetics Conference Asilomar 17, Pacific Grove, CA*. Genetics Society of America, 2017.

Poster presentations

2025 **Hyunbin Kim**, Rachel Seongeun Kim, Milot Mirdita, and Martin Steinegger. Unraveling protein structural motifs across the protein universe with folddisco. In *RECOMB 2025, 29th Annual International Conference on Research in Computational Molecular Biology, Seoul, Korea*. ISCB, 2025.

2024 **Hyunbin Kim**, Rachel Seongeun Kim, Milot Mirdita, and Martin Steinegger. Structural motif detection on the scale of the protein universe with folddisco. In *APBJC2024, 1st Asia&Pacific Bioinformatics Joint Conference JSBi, GIW, InCoB, APBC, and ISCB-Asia, Naha, Japan*. ISCB, 2024.

2023 **Hyunbin Kim**, Milot Mirdita, and Martin Steinegger. Foldcomp: scalable solution for compressing huge protein structure database. In *ISMB/ECCB 2023, The 31st Annual Intelligent Systems For Molecular Biology and the 22nd Annual European Conference on Computational Biology, Lyon, France*. ISCB, 2023.

2022 **Hyunbin Kim**, Johannes Söding, and Martin Steinegger. Fast lossy protein structure compression algorithm. In *ISMB 2022, The 30st Annual Intelligent Systems For Molecular Biology, Madison, Wisconsin*. ISCB, 2022.

2017 **Hyunbin Kim** and Yong-Hwan Lee. Deciphering protein-protein interaction network in the rice blast fungus, *Magnaporthe oryzae*. In *2017 APS Annual Meeting, San Antonio, Texas*. American Phytopathological Society, 2017.

2017 **Hyunbin Kim**, K. Cheong, K.T. Kim, J. Jeon, G. Choi, and Y.H. Lee. Magnet: the integrated gene network of the rice blast fungus *Magnaporthe oryzae*. In *29th Fungal Genetics Conference Asilomar 17, Pacific Grove, California*. Genetics Society of America, 2017.

Teaching Assistant

Fall, 2021 **Advanced Bioinformatics**, Seoul National University.
Feedback to the student presentation materials for improvement

Spring, 2021 **Introduction to Bioinformatics**, Seoul National University.
Assignment & exam design

Certificate

2019 **Certificate of competency in Fundamental of Deep Learning for Natural Language**, *NVIDIA Deep Learning Institute*, Seoul, South Korea.